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FLAT

Intro: Mathematical computations, mathematical models.

(singular)

Automata: It is a self acting device

Automata Theory: It is a study of abstract computing devices / Machines. [based on input → languages finite]

set of states

Finite Automata: it is a useful kind of hardware / software

Applications →
1. used for checking behaviour / software of digital circuits
2. lexical
3. Scanning of large bodies of text (web page)
4. int a; ✓ int a, x
5. Communication Protocols.

Symbols: a, b, 0, 1, . . .

Alphabet: {0, b}; {a, b, c}, {0, 1}, {0, 1, . . . 9} — 'Σ' / (W)

String: ε (empty), a, b, aa, ab, ba, bb, aaa, bbb, abb . . . . n

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Language : Collection of Strings $\{a, b\} \rightarrow 5$ → let max. length of string

$L_1 = \{\epsilon, a, b, aa, bb, ab, ba\} \rightarrow$ Finite

$\Sigma = \{a, b\}$ L_2 Set of all strings of length = only $3 \rightarrow \{aaa, bbb, \ldots 8\} \rightarrow$ Finite

$\Sigma = \{a, b\}$ L_3 = String starting with a

$\{a, aa, ab, aaa, aab \ldots \infty\} \rightarrow$ Infinite

Power of Σ :

$\Sigma^0 = |\epsilon| = 0$

$\Sigma = \{a, b\}$ $\Sigma^1 = \{a, b\}$

$\Sigma^2 = \Sigma^1 \Sigma^1 = \{aa, ab, ba, bb\}$

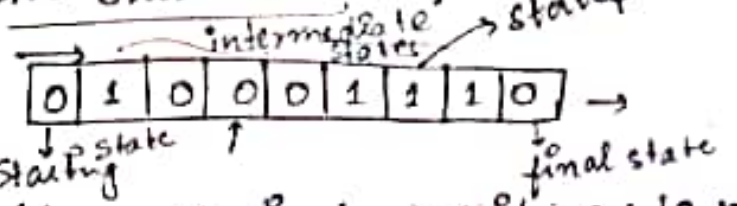
$\Sigma^3 = \Sigma \Sigma . \Sigma = \{a, b\}\{a, b\}\{a, b\}$

Σ^n = n length

$\Sigma^* = \Sigma^+$

kleen closure +ve closure.

Finite State Machine :



If match is found it goes to next

Else it gives that match is not found

Ex: for 3

0 1 2 3 4 5 → dead states

final state

Check whether all states are passed and then goes to the last

Ex: for 3

0 1 2 3 →

Finite Automata → with o/p → moore machine, mealy machine
→ without o/p → Deterministic FA, NDFA (Non), Epilson NDFA

FA: $(Q, \Sigma, \delta, q_0, F)$ ← [5 tuple definition]

$Q \rightarrow$ represent set of states : $\{A, B, C\}$

Ex: $\rightarrow (A) \xrightarrow{a} (B) \xrightarrow{b} (C)$

$\Sigma : \{a, b\}$ Input symbols

q_0 : Initial state $\{A\}$

F : Final State (Double circle) : $\{C\}$
(acceptance state)

$Q \supseteq F$

δ : Transition States.

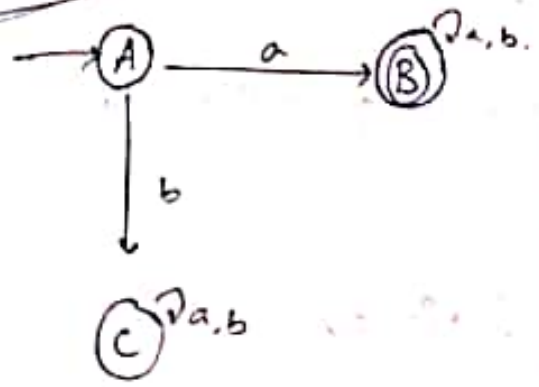
$Q = Q \times \Sigma$

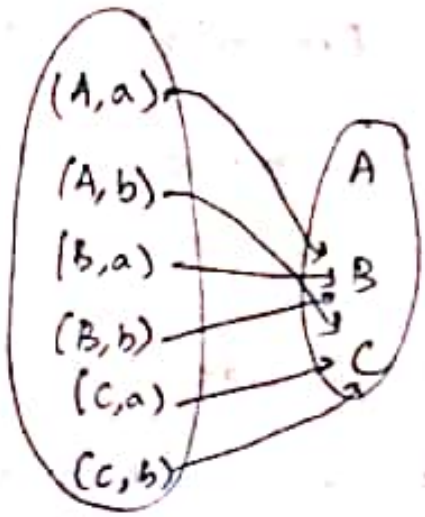
$Q = \{A, B, C\} \times \{a, b\}$

$Q = (A, a)\ (A, b)\ (B, a)\ (B, b)\ (C, a)\ (C, b)$

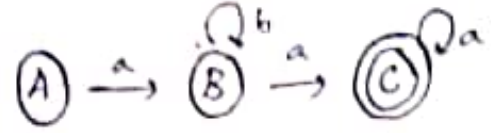
δ	a	b
A	B	C
B	B	B
C	C	C

DFA





$\phi \rightarrow$ If we have no state in transition diagram.

Ex: 

δ	a	b
A	B	ϕ
B	C	B
C	C	ϕ

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$\Sigma^+ = \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \ldots$ [Positive combinations]

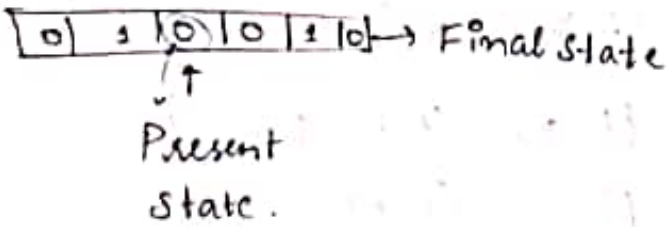
$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \ldots$

$*$ → means many

Kleen closure.

Finite automata

Set of states



DFA → Deterministic FA

At can mo only 1 state for each input ~~signal~~ symbol.

NDFA → Non-Deterministic FA

At can move to more than 1 state also for a input ~~signal~~ symbol.

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* DFA

1. A finite set of states denoted Q.
2. A finite of set of input symbols Σ.
3. transition function δ.

   $q \rightarrow$ current state

   $a \rightarrow$ input signal symbol $[a \in \Sigma]$

   $p \rightarrow$ next state $[q, p \in Q]$

   $\boxed{\delta(q, a) = p}$ [Travels from q to p with label a]

4. A start state, one of the state in Q (q_0)
5. A set of final / accepting states, F $[F \in Q]$.

$\boxed{A = (Q, \Sigma, \delta, q_0, F)}$ → Formal def. of DFA.

(name of DFA)

Let $a_1, a_2, a_3, \ldots, a_n$ → input symbols

$q_0 \rightarrow$ initial/Start state

$\delta(q_0, a_1) = q_1$

$\delta(q_1, a_2) = q_2$

$\delta(q_2, a_3) = q_3$

$\vdots$

$\delta(q_{i-1}, a_i) = q_i \quad \forall i$

q_n

If $\boxed{q_n \notin F}$ then string is not accepted by DFA → meaning → machine does not reach the Final state.

If $\boxed{q_n \in F}$ input symbol string is accepted by DFA, meaning → the machine reaches Final State.

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01 → if present in string, known as string of lang.

Ex: {01, 001, 011, 0111, 1101}

{ w/w is from the x01y for some strings x & y consists of 0's and 1's }

Ex: { x0y / x & y are any string of 0's and 1's }

→ any string starting with 0. $\Sigma = \{0, 1\}$

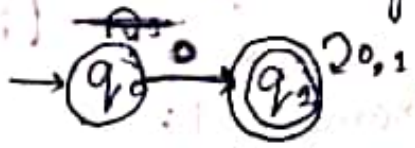
$L = \{0, 01, 011, 001, \ldots\}$ [If $\delta(q_2, 1) = q_2$ → dead state since shld start with 0]

$\delta(q_0, 0) = q_1$ [∵ string should start with 0, for 1st δ state changes, next regardless of input symbol state doesn't change.]

$\delta(q_1, 0) = q_1$

$\delta(q_1, 1) = q_1$

⇒ q_1 = Final state



⇒ $A = (\{q_0, q_1\}, \{0, 1\}, \delta, q_0, \{q_1\})$

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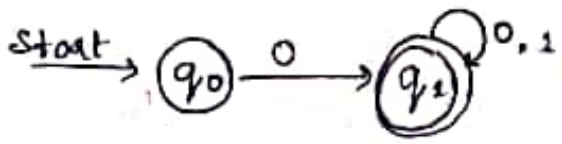

$\delta(q_0, 0) = q_1$

$\delta(q_1, 0) = q_1$

$\delta(q_1, 1) = q_1$

δ	0	1
→ q_0	q_1	ϕ
q_1	q_1	q_1

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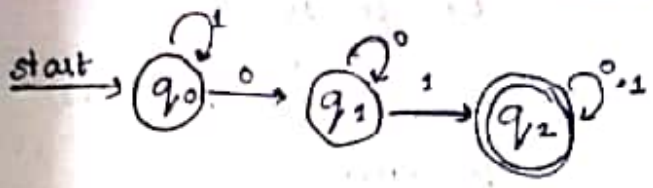
→ Language ending in 01

$\Sigma = \{0, 1\}$

$L = \{01, 001, 101, 0001, 0101, 1001, 1101, \ldots\}$



$\delta(q_0, 0) = q_1$

$\delta(q_0, 1) = q_0$

$\delta(q_1, 1) = q_2$

$\delta(q_1, 0) = q_1$

$\delta(q_2, 0) = q_2$

$\delta(q_2, 1) = q_2$ } (*)

	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_1	q_2
q_2^*	q_2	q_2

Ex: for $\boxed{0001}$

$q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_1 \xrightarrow{0} q_2 \xrightarrow{1} q_2$

0001

$\delta(q_0, 0001) = \delta(\delta(q_0, 0), 001)$

$= \delta(q_1, 001)$

$= \delta(\delta(q_1, 0), 01)$

$= \delta(q_1, 01)$

$= \delta(\delta(q_1, 0), 1)$

$= \delta(q_1, 1)$

$= q_2$

$D(A) = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_2\})$

Here δ is defined as (*)

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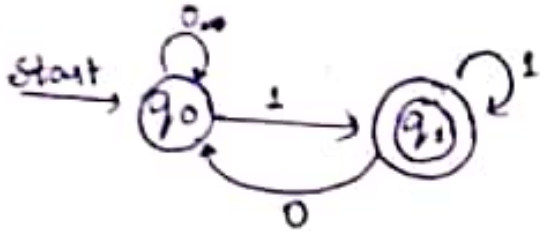
→ Langugge ending with 1. [or for odd numbers [binary odd]]

L = {1, 01, 11, 001, 011, 101, 111, ...}



Ex: for 101

$q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_0 \xrightarrow{1} q_1$

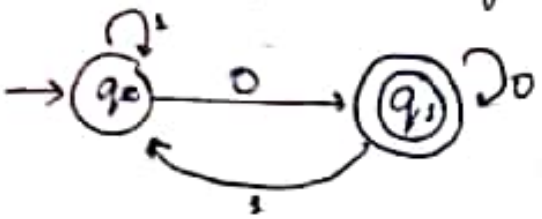
for 1101

$q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_1 \xrightarrow{0} q_0 \xrightarrow{1} q_1$

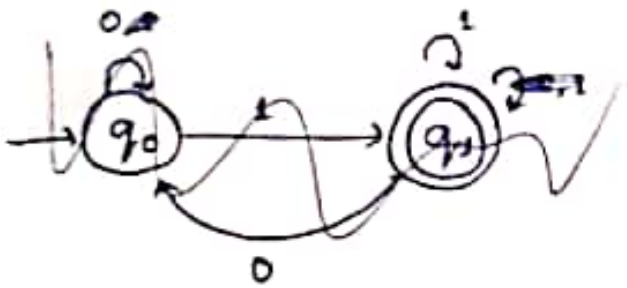
	0	1
q_0	q_0	q_1
q_1	q_0	q_1

→ Design a FA which accepts odd number of 1's and any number of 0's.

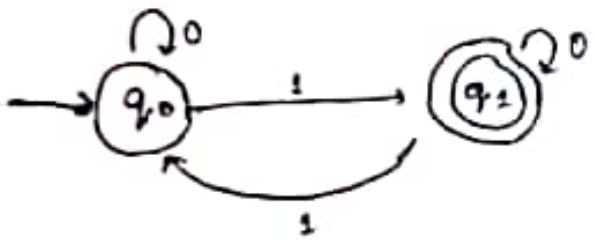
→ Language ending with 0. [even number]



$q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1$



$q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_1$



→ odd no. of 1's & ~~even~~ any no. of 0's

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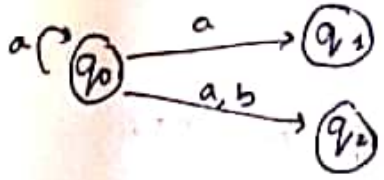
* Non-Deterministic FA:

$\delta = Q \times \Sigma \rightarrow 2^Q$

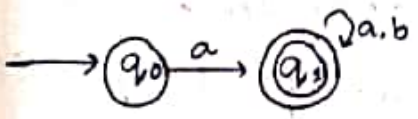
- No concept of deadstates.
- Can convert NFA to DFA.



Ex: $\Sigma = \{a, b\}$

Language starts with a

$L = \{a, aa, ab, aab, a \ldots\}$

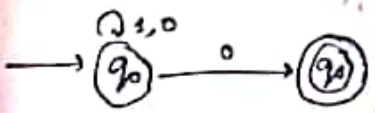


→ DFA

	a	b
$\rightarrow q_0$	q_1	ϕ
q_1^*	q_1	q_1

2. $\Sigma = \{0, 1\}$ ends with 0

$L = \{0, 010, 00, 1010, 000, \ldots\}$



$\delta(q_0, 0010) = \delta(q_0, 010)$

$= \delta(q_0, 10)$

$= \delta(q_0, 0)$

$= q_1$

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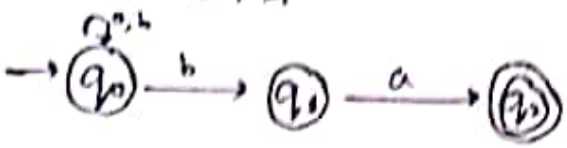
* L = {ba, aba, bba, aaba, ...} Ending with ba

* NFA & DFA

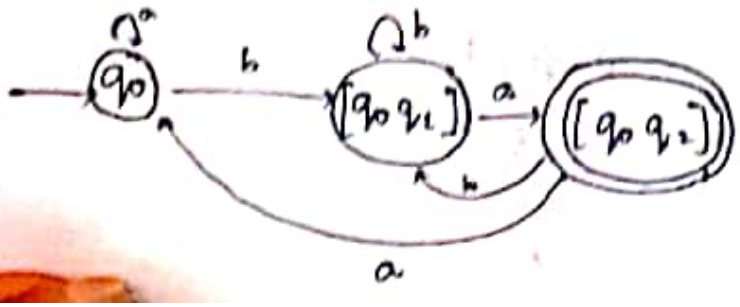
1. Initial state is same as NFA only for new DFA
2. $\delta(q_0, q_1, \ldots, q_n, x) = \bigcup_{i=1}^{n} \delta(q_i, x)$
3. Every subset that contains the final state of NFA is a final state of DFA.
4. Start construction of δ from the initial state and continue the process for every new state that comes under I/P column. Terminate the process whenever no new state appear under I/P column.

NFA → DFA



NFA	a	b
→ q_0	q_0	$\{q_0, q_1\}$
q_1	q_2	ϕ
q_2	ϕ	ϕ

DFA	a	b
→ q_0	q_0	$[q_0 q_1]$
$[q_0 q_1]$	$[q_0 q_2]$	$[q_0 q_1]$
$[q_0 q_2]$	q_0	$[q_0 q_2]$



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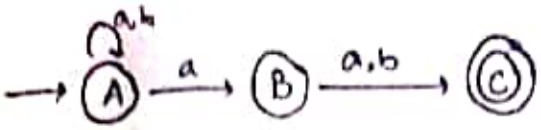


NFA	a	b
→ A	[AB]	A
B	C	C
C*	∅	∅

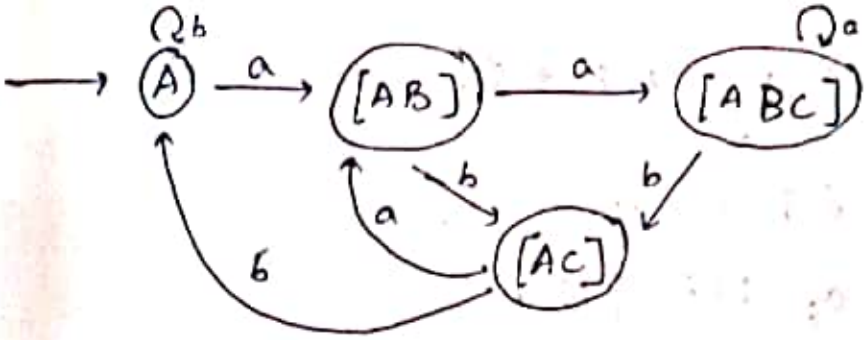
DFA	a	b
→ A	[AB]	A
[AB]	[ABC]	[AC]
[ABC]	[ABC]	[AC]
[AC]	[AB]	A

$\delta(A(a)) \cup \delta(B(a))$

$= [AB] \cup [C]$

$= [ABC] \cup \phi = [ABC]$

$[AC] \cup \phi = [AC]$



→

δ	0	1
→ p	{pq}	{p}
q	∅	{r}
r*	{pr}	{q}

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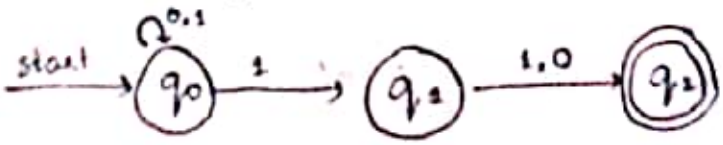
→ Design NFA which accepts exactly those strings that have the symbol 1 in second last position.



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$\delta(q_0, 1) = \{q_0 q_1\}$

$L_N(A) = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_2\})$

δ is defined as follows.

	0	1
$\rightarrow q_0$	q_0	$q_0 q_1$
q_1	q_2	q_2
q_2	ϕ	ϕ

$A = (Q, \Sigma, \delta, q_0, F)$ is an NFA then

$L(A) = \{w / \delta(q_0, w) \cap F \neq \phi\}$

or

$\{w / \delta(q_0, w)$ is in $F\}$

<u>Equivalence</u> of <u>DFA</u> & <u>NFA</u>

Every DFA is NFA but every NFA is not DFA.

$N = (Q_N, \Sigma, \delta_N, q_0, F_N)$

$D = (Q_D, \Sigma, \delta_D, q_0, F_D)$

Q_D is the set of subset of Q_N.

Q_D is the power set of Q_N

If Q_N has n states then Q_D will have 2^n states.

F_D is the set of subsets of Q_N

$\delta \cap F_N \neq \phi$

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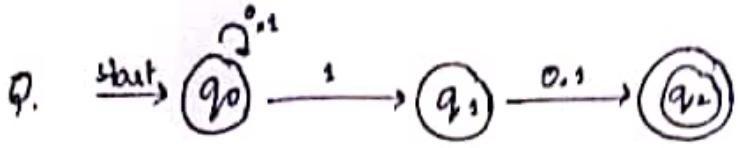
$\delta \subseteq Q_N$ and for each input symbol a in Σ

$$\delta_D(S, a) = \bigcup \delta_D(p, a)$$

Q.



	0	1
$\rightarrow q_0$	q_0	$q_0 q_1$
q_1	q_2	q_2
q_2^*	ϕ	ϕ

The possible states in DFA are

ϕ, $\{q_0\}$, $\{q_1\}$, $\{q_2\}$, $\{q_0, q_1\}$

$\{q_0, q_2\}$, $\{q_1, q_2\}$, $\{q_0, q_1, q_2\}$

Final states in DFA

$\{q_2\}$ $\{q_0 q_2\}$ $\{q_1, q_2\}$, $\{q_0, q_1, q_2\}$

$\delta(q_0, 0) = q_0$

$\delta(q_0, 1) = \{q_0, q_1\}$

$\delta(q_1, 0) = q_2$

$\delta(q_1, 1) = q_2$

$\delta(q_2, 0) = \phi$

$\delta(q_2, 1) = \phi$

$$\delta(\{q_0, q_1\}, 0) = \delta(q_0, 0) \cup \delta(q_1, 0)$$
$$= \{q_0\} \cup \{q_2\}$$
$$= \{q_0, q_2\}$$

$$\delta((q_0, q_1, q_2), 1) = \delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1)$$
$$= \{q_0, q_1\} \cup \{q_2\} \cup \phi$$
$$= \{q_0, q_1, q_2\}$$

Solve for all 8 possible states in this way and write the transaction table and diagram.

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* Moore Machine [o/p depends only on present state]

$(Q, \Sigma, \Delta, \delta, \lambda, q_0)$

$Q \rightarrow$ finite set of sets

$\Sigma \rightarrow$ finite set of input signals

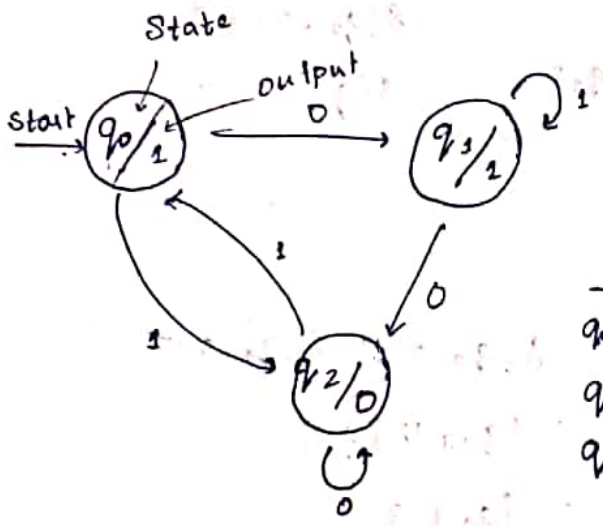
$\Delta \rightarrow$ output alphabet

$\delta \rightarrow$ transition function $Q \times \Sigma \rightarrow Q$

$\lambda \rightarrow$ is o/p fn. such that $Q \rightarrow \Delta$

$q_0 \rightarrow$ start state.

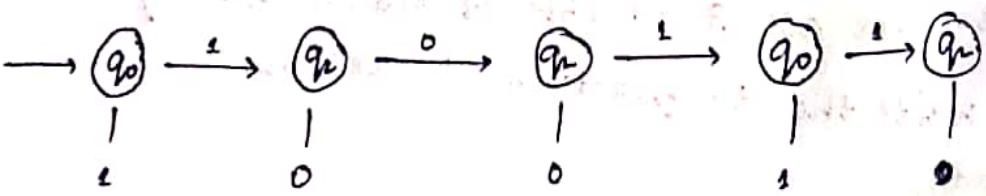
Ex:



	0	1	o/p
q_0	q_1	q_2	1
q_1	q_2	q_1	1
q_2	q_2	q_0	0

$\boxed{\begin{array}{l} n \text{ - bits input} \\ \text{Output: } (n+1) \text{ bits.} \end{array}}$

Ex: i/p $\rightarrow$ 1011.



o/p $\rightarrow$ 10010

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* **Mealy Machine** [o/p depends on present state & i/p signal].

$(Q, \Sigma, \Delta, \delta, \lambda, q_0)$

$Q \rightarrow$ finite set of states

$\Sigma \rightarrow$ finite set of input symbols.

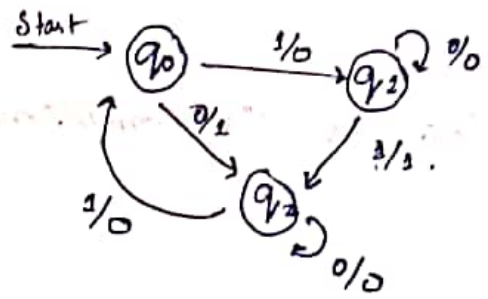
$\Delta \rightarrow$ o/p alphabet

$\delta \rightarrow$ transition fn. $Q \times \Sigma \rightarrow Q$

$\lambda \rightarrow$ output fn. $Q \times \Sigma \rightarrow \Delta$

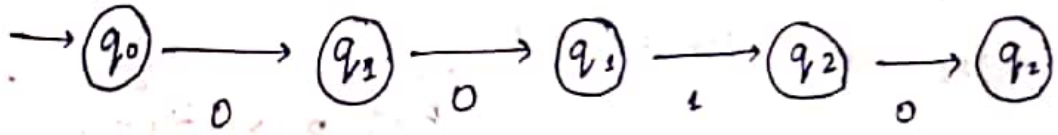
$q_0 \rightarrow$ start state

Ex:



	0	o/p	1	o/p
q_0	q_2	1	q_1	0
q_1	q_1	0	q_2	1
q_2	q_2	0	q_0	0

Ex: i/p : 1 0 1 0



$\rightarrow$ o/p : 0 0 1 0

i/p $\rightarrow$ n bits
o/p $\rightarrow$ n bits

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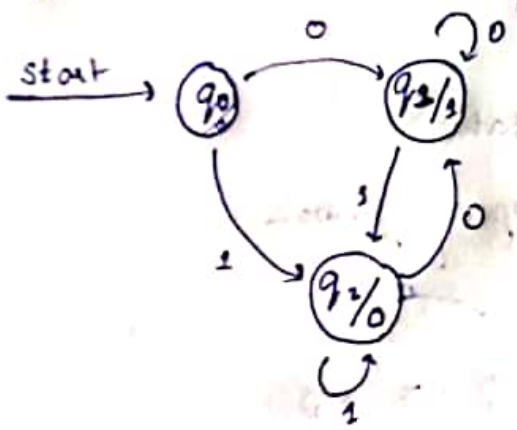
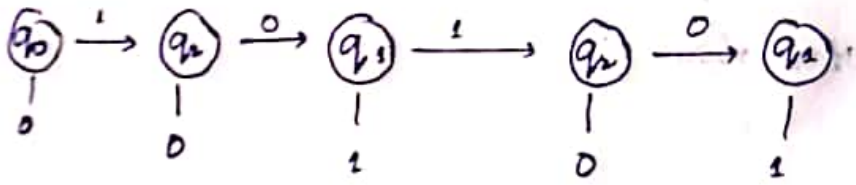
→ Design a moore machine for 1's complement of binary number.



	0	1	o/p
q_0	q_1	q_2	0
q_1	q_1	q_2	1
q_2	q_1	q_2	0

i/p: 1010



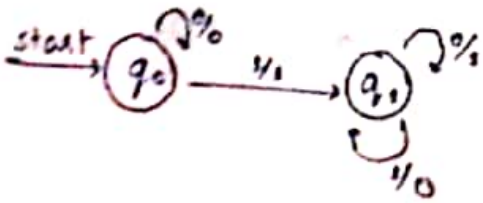
Ans: 00101 → o/p

→ Design a mealy machine for 2's complement of binary number.

1's complement + 1 = 2's complement

From right the digits are same till 1st 1. From the rightmost 1 towards left digits are interchanged.

Ex: i/p → 1010 0
0 1 1 0 0 → o/p



We are going right to left.

	0	o/p	1	o/p
q_0	q_0	0	q_1	1
q_1	q_1	1	q_1	0

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